Video 04 | Filters and the Insert

Massive PSXL instructional series - video script

Runtime estimate	Narration length	Scenes	Format
about 5:31 minutes	782 words at 150 words per minute	12	Screen capture of the plug-in with voice-over; dark theme, interface size 125 percent

Learning goals

- Choose between the four filter models and their modes.
- Use Cutoff, Reso, Drive and Key Track, and chain two filters.
- Use the Insert's six types for colour and distortion.

NOTE. Accuracy rule for the narrator and any generated video: use the on-screen names exactly as written in the action column. After **Load Init (Blank)** the synth is silent until routing is connected. Do not compare Massive PSXL to other products.

Scene table

Scene	On-screen action	Narration (word for word)	Visual notes and call-outs
1	Init (Blank), quick setup Osc 1 + Osc 2 > Filter 1 > Out; Osc 1 Pos 0.66, Osc 2 table Morph Basic Pos 0.66, Unison 3 both.	We need a bright source to hear a filter, so build a quick patch: Osc 1 plus Osc 2 into Filter 1 into Out, both on the saw position of Morph Basic, with a little unison. Saws are full of harmonics, which gives the filter something to remove. Listening to a filter on a sine wave would teach us very little.	Show the routing with the glowing wires.
2	Filter tab Filter 1: Model Ladder, Mode LP 24. Sweep Cutoff slowly; then raise Reso to 0.6 and sweep again.	The Ladder model is the classic fat low-pass. Sweep Cutoff and the sound closes down. Raise Reso for a pronounced peak. Near the top of the range the filter nearly rings like a sine. LP 24 is four poles; LP 12 is gentler; BP keeps a band; and HP 24 removes the low end. While sweeping, listen to how the resonance peak moves up and down through the harmonics. The bigger the peak, the more vocal and nasal the sound becomes. Back off the resonance if the sound gets harsh.	Cycle through the four modes with their names visible in the Mode menu.
3	Switch Model to State Variable; click through LP, BP, HP, Notch while sweeping.	State Variable is a clean twelve decibel filter with four modes: low-pass, band-pass, high-pass and notch. The notch is great for phaser-like sweeps. The state variable filter is a little cleaner and more precise than the ladder, which makes it good for drums and for effects where you want a tidy, controlled response.	Spectrum-style graphic overlay for each mode (simple curves).
4	Switch Model to Comb; set Cutoff 300 Hz, Reso 0.6, KeyTrk +1 and play a few notes; cycle Comb +, Comb -, Flange, Pluck.	Comb is a tiny feedback delay. Here Cutoff sets the pitch of the resonance. Turn Key Track up to plus one and the comb follows your keyboard. Comb plus rings with all harmonics, Comb minus is hollow, Flange sweeps notches, and Pluck gives a string-like resonance. A comb filter works best on short or noisy sounds, and on sources with a lot of harmonics. Experiment with different cutoff values until the resonance matches the notes you play.	Highlight KeyTrk knob when it is turned.

Scene	On-screen action	Narration (word for word)	Visual notes and call-outs
5	Switch Model to Formant; Mode Male; Reso 0.5; sweep Cutoff from low to high slowly; then Female, Child, Alien.	Formant turns the filter into a vowel machine. Cutoff now chooses the vowel, A at the bottom through to U at the top, and Reso sharpens it. Male, Female, Child and Alien change the voice type. Between the vowels the filter interpolates smoothly, so sweeping the cutoff with an LFO or envelope makes the sound speak. The vowel positions are roughly one hundred and ten hertz for E, six hundred thirty for I, and three and a half kilohertz for O.	Overlay A E I O U along the Cutoff sweep.
6	Back to Ladder LP 24; raise Filter 1 Drive from 0 to 0.7 and back.	Drive pushes the signal into saturation before the filter, which fattens and roughens. It is different from Amp Feedback, which loops the whole voice back into Filter 1. Both are worth trying. Think of Drive as a way to add warmth and bite before the filter. Because the filter comes after, the saturation gets shaped, so a closed filter turns a harsh drive into a pleasant growl.	Hover values for Drive; short callout for 'Amp Feedback' knob in the Amp panel.
7	Routing: add Filter 1 > Filter 2 > Out (series). Set Filter 2 to Model State Variable, Mode HP, Cutoff 200 Hz.	Two filters are better than one. Wire Filter 1 into Filter 2 and then to Out. Make Filter 2 a high-pass at two hundred hertz, and you have carved a band between them. Series gives steeper shapes; for parallel use the quick setup and send each oscillator to its own filter. Filter 2 has its own model, mode, cutoff, resonance, drive and key track, so the two filters can be completely different. Many classic sounds come from a low-pass and a high-pass working together.	Show both Filter tabs in sequence and the quick setup list.
8	Raise Filter 1 KeyTrk to +0.5 and play across the keyboard.	Key Track makes the cutoff follow the note. At plus one it moves one octave per octave, so bright and dark notes stay balanced across the keyboard. Key Track only changes where the cutoff sits relative to middle C. Use negative values if you want low notes to be brighter than high notes, which is unusual but can be fun.	Short audio demo low-mid-high with the knob at 0 then +0.5.
9	Routing: Filter 2 > Insert > Out. Insert Type TanH, Drive 0.5, Mix 0.7.	Now the Insert. It is a per-voice shaper that sits in the routing like a node. Wire a filter into it and out to Out. TanH is smooth saturation. The insert only does something if a wire enters it. If you ever add the insert and hear nothing, check that it is connected on both sides.	Highlight the amber Insert node and its panel.
10	Step through Insert types: Fold, Bitcrush, Hard Clip, Rectify, Sine; adjust Drive and Mix for each.	Fold is metallic. Bitcrush reduces bit depth and sample rate for lo-fi grit. Hard Clip is aggressive. Rectify flips the negative half for an octave-up buzz. Sine is smooth at low drive and glassy at high drive. Mix blends the effect with the dry signal. Drive controls how hard the signal is pushed, and mix blends the processed sound with the original. Both can be modulated, so try an envelope on Insert Drive for a bright, crunchy attack that smooths out.	Pause on each name for 2 seconds with a short sound demo.
11	Insert: Mix 0.5. Show Amp panel Level, Pan, Feedback; sweep Pan. Show Volume knob in the top bar.	After the Insert comes the Amp. Level, Pan and Feedback. Feedback adds grit. The master Volume sits in the top bar and comes after the effects. Amp Pan is equal-power, so the level stays consistent as you move across the stereo field. Modulating pan with an LFO gives you auto-pan.	Callout: voice level = amp envelope x Level x velocity.
12	End on the full patch with a held chord.	You now know the filter models and the insert. In the next video we make these controls move, with envelopes and LFOs.	End card: 'Next: 05 Envelopes and LFOs'.

Narration only (plain script for text-to-speech or NotebookLM)

[Scene 1] We need a bright source to hear a filter, so build a quick patch: Osc 1 plus Osc 2 into Filter 1 into Out, both on the saw position of Morph Basic, with a little unison. Saws are full of harmonics, which gives the filter something to remove. Listening to a filter on a sine wave would teach us very little.

[Scene 2] The Ladder model is the classic fat low-pass. Sweep Cutoff and the sound closes down. Raise Reso for a pronounced peak. Near the top of the range the filter nearly rings like a sine. LP 24 is four poles; LP 12 is gentler; BP keeps a band; and HP 24 removes the low end. While sweeping, listen to how the resonance peak moves up and down through the harmonics. The bigger the peak, the more vocal and nasal the sound becomes. Back off the resonance if the sound gets harsh.

[Scene 3] State Variable is a clean twelve decibel filter with four modes: low-pass, band-pass, high-pass and notch. The notch is great for phaser-like sweeps. The state variable filter is a little cleaner and more precise than the ladder, which makes it good for drums and for effects where you want a tidy, controlled response.

[Scene 4] Comb is a tiny feedback delay. Here Cutoff sets the pitch of the resonance. Turn Key Track up to plus one and the comb follows your keyboard. Comb plus rings with all harmonics, Comb minus is hollow, Flange sweeps notches, and Pluck gives a string-like resonance. A comb filter works best on short or noisy sounds, and on sources with a lot of harmonics. Experiment with different cutoff values until the resonance matches the notes you play.

[Scene 5] Formant turns the filter into a vowel machine. Cutoff now chooses the vowel, A at the bottom through to U at the top, and Reso sharpens it. Male, Female, Child and Alien change the voice type. Between the vowels the filter interpolates smoothly, so sweeping the cutoff with an LFO or envelope makes the sound speak. The vowel positions are roughly one hundred and ten hertz for E, six hundred thirty for I, and three and a half kilohertz for O.

[Scene 6] Drive pushes the signal into saturation before the filter, which fattens and roughens. It is different from Amp Feedback, which loops the whole voice back into Filter 1. Both are worth trying. Think of Drive as a way to add warmth and bite before the filter. Because the filter comes after, the saturation gets shaped, so a closed filter turns a harsh drive into a pleasant growl.

[Scene 7] Two filters are better than one. Wire Filter 1 into Filter 2 and then to Out. Make Filter 2 a high-pass at two hundred hertz, and you have carved a band between them. Series gives steeper shapes; for parallel use the quick setup and send each oscillator to its own filter. Filter 2 has its own model, mode, cutoff, resonance, drive and key track, so the two filters can be completely different. Many classic sounds come from a low-pass and a high-pass working together.

[Scene 8] Key Track makes the cutoff follow the note. At plus one it moves one octave per octave, so bright and dark notes stay balanced across the keyboard. Key Track only changes where the cutoff sits relative to middle C. Use negative values if you want low notes to be brighter than high notes, which is unusual but can be fun.

[Scene 9] Now the Insert. It is a per-voice shaper that sits in the routing like a node. Wire a filter into it and out to Out. TanH is smooth saturation. The insert only does something if a wire enters it. If you ever add the insert and hear nothing, check that it is connected on both sides.

[Scene 10] Fold is metallic. Bitcrush reduces bit depth and sample rate for lo-fi grit. Hard Clip is aggressive. Rectify flips the negative half for an octave-up buzz. Sine is smooth at low drive and glassy at high drive. Mix blends the effect with the dry signal. Drive controls how hard the signal is pushed, and mix blends the

processed sound with the original. Both can be modulated, so try an envelope on Insert Drive for a bright, crunchy attack that smooths out.

[Scene 11] After the Insert comes the Amp. Level, Pan and Feedback. Feedback adds grit. The master Volume sits in the top bar and comes after the effects. Amp Pan is equal-power, so the level stays consistent as you move across the stereo field. Modulating pan with an LFO gives you auto-pan.

[Scene 12] You now know the filter models and the insert. In the next video we make these controls move, with envelopes and LFOs.

> **TIP.** Pause for about one second between scenes and let on-screen actions finish before the next sentence starts.